

Physiological, Aerodynamic, and Acoustic Differences between Male and Female Voices

QUICK REVIEW

- The vocal tract of an adult female is about 15% shorter than an adult male's
- This can be attributed to the disproportionally longer pharynx that adult males have in comparison to females
- A man's larynx is about 20% larger than that of women and the length of the membranous part of the vocal folds differ by about 60%

F0: DIFFERENCES BETWEEN MEN AND WOMEN

- Fundamental frequency remains a foolproof way to distinguish between male and female voices owing to physiological differences
- The average F0 of men and women differ by about an octave
- The mean F0 of male voices is about 130 Hz and that of female voices is 220 Hz
- F0 differences allow for discerning male voice from female voice with a 96% accuracy
- The differences in F0 is contingent on vocal fold length

HORMONAL INFLUENCE ON LARYNX AND VOICE

- The mammalian larynx is a hormonal target organ
- Sex hormone levels significantly affect laryngeal function throughout life
- Testosterone determines laryngeal size and vocal fold length during puberty
- Absence of androgens leads to female voice characteristics
- Presence of androgens causes male vocal characteristics
- Higher testosterone levels are associated with lower-pitched voices in males

HORMONAL INFLUENCES ON WOMEN

- Menopause: Decreased estrogen can thin vocal fold mucosa
- Menstrual cycle: Potential subtle changes in voice
- Possible increases in mucous secretions
- Potential vocal fatigue, though mostly imperceptible
- Oral contraceptives show no significant impact on voice quality
- Virilizing agents can permanently lower women's vocal pitch
- Pregnancy's only noted vocal effect: decreased maximum phonation duration

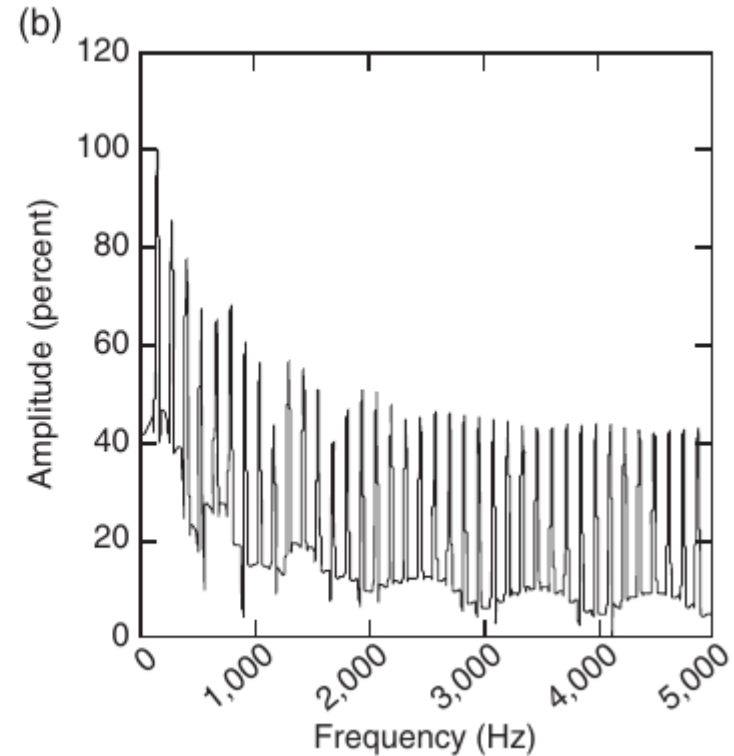
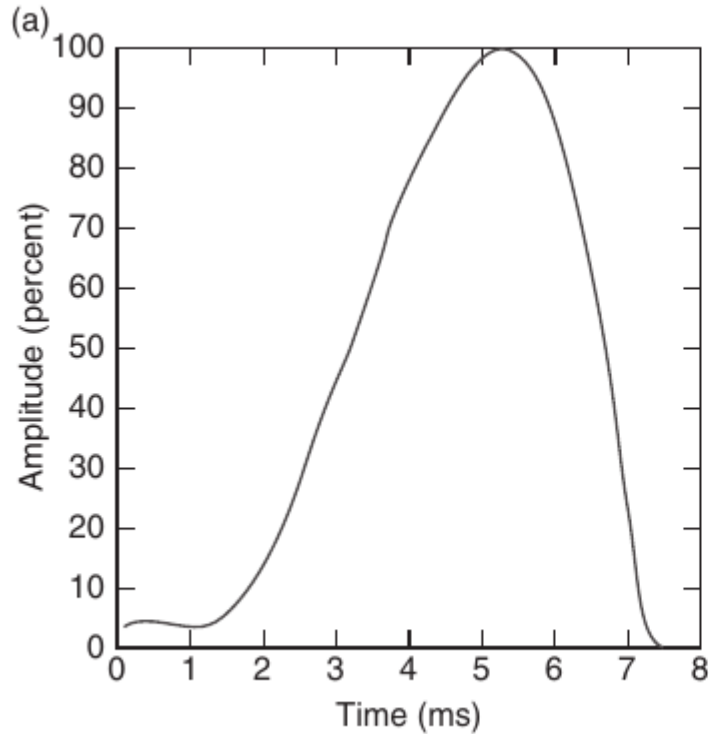
VOCAL FOLDS

- Although the structure of the vocal folds remain the same across both sexes, the differences pertaining to the vocal folds lie in the pattern and rate of vibration of the vocal folds
- Women's vocal folds open and close smoothly, in a quasi-sinusoidal pattern
- Men's vocal folds, on the other hand, close more quickly than they open
- The glottal pulses of male and female voices reflect the same

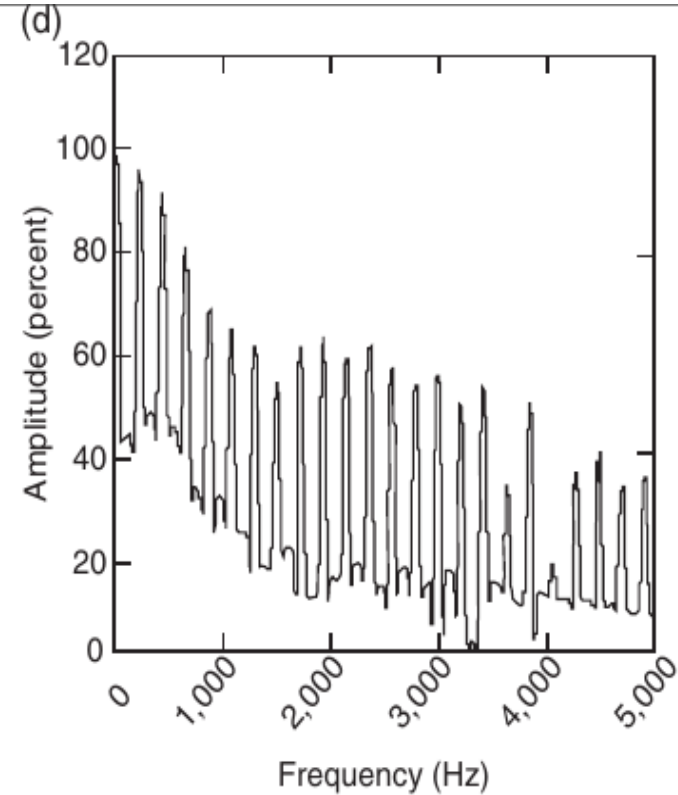
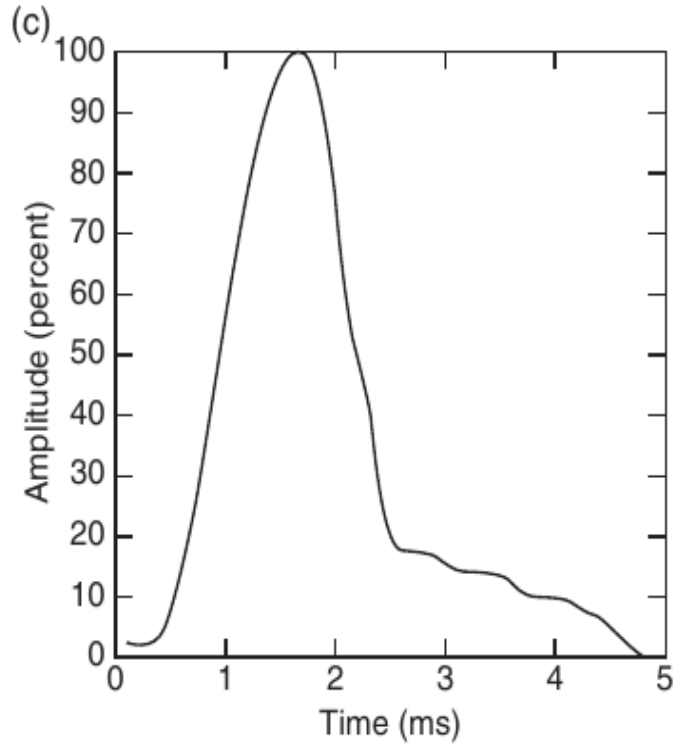
Voices – Source Pulse and Spectrum (Talking Point)

- The Male Pulse closes more sharply
- Female Pulses are a bit more symmetrical
- The lack abrupt discontinuity in airflow when the vocal folds vibrate sinusoidally, results in female voices being characterised by less high frequency acoustic energy than male voices, indicated by lower harmonic heights in the spectra
- Female voices are considered breathier and weaker owing to the aspiration caused by the persistent glottal gap that is observed in women throughout phonatory cycles.

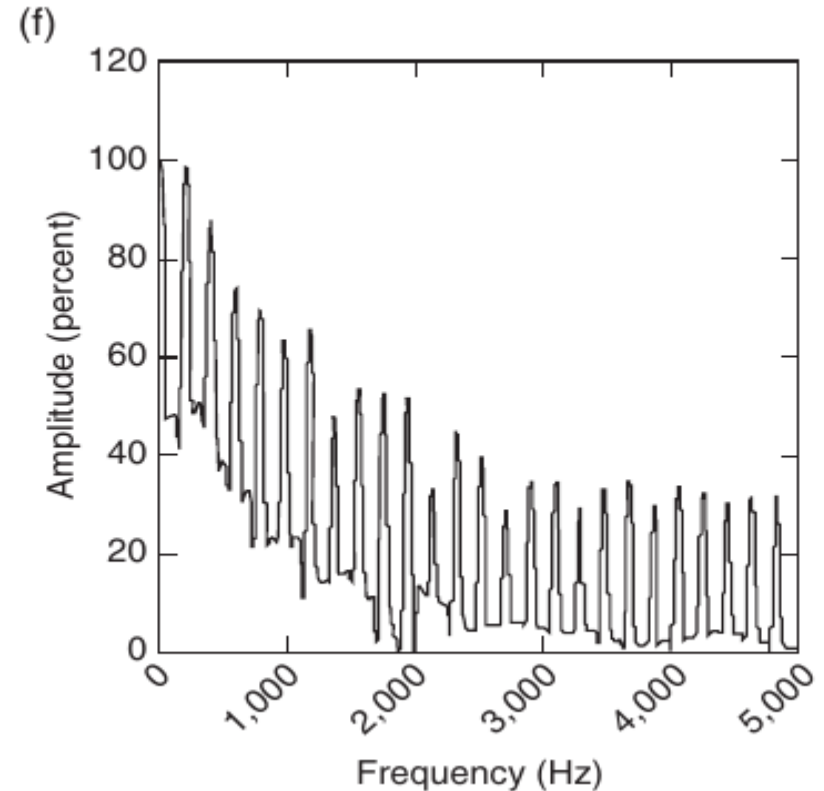
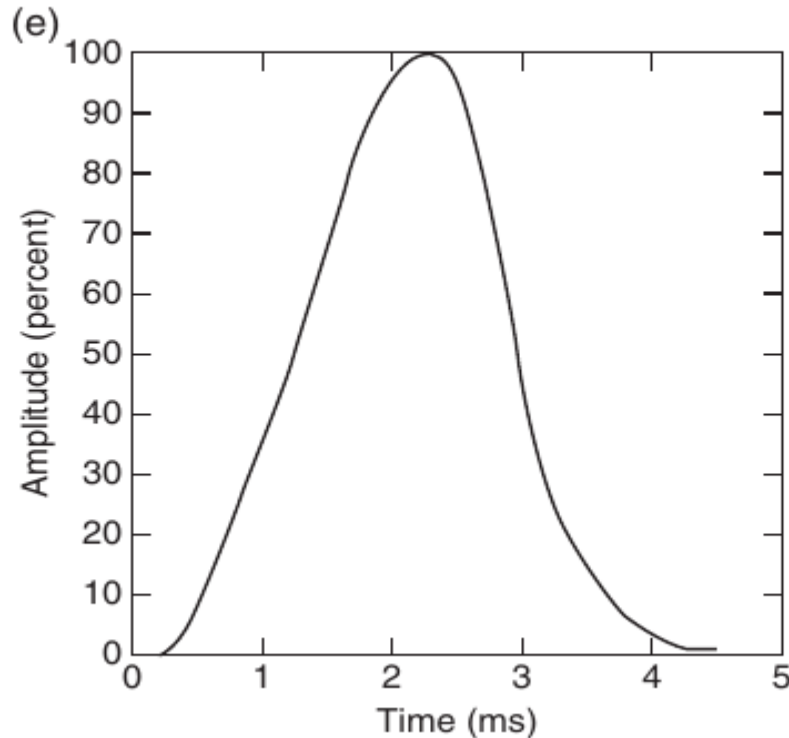
Male Voice – Source Pulse and Spectrum



Female Voice – Source Pulse and Spectrum (Efficient)



Female Voice – Source Pulse and Spectrum (Breathy)



Vocal Source and Frequency Variation

- Significant variability exists in voice production within and across genders
- Women's intonation is often described as more "dynamic"
 - Wider pitch range
 - Sharper pitch changes
 - More animated speaking style

Vocal Source and Frequency Variation -Talking Points

- **Additional Talking Points:**
- Not all studies agree on pitch differences
- Some researchers suggest pitch variability depends on the speaking task
- Individual variation is more significant than broad gender generalizations
- The perception of pitch can differ based on measurement scales (Hertz vs. semitone)

Pronunciation and Linguistic Precision

- Women tend to have more "precise" pronunciation
 - More accurate individual sound production
 - Less likely to omit sounds (e.g., "-ing" endings)
 - Generally more intelligible speech
- Findings consistent across multiple languages and cultures
- Pronunciation precision can vary by context and individual

Common Misconceptions and Actual Speech Patterns

- Contrary to popular belief:
 - Men and women speak approximately the same number of words daily (16,000)
 - Speaking rates are similar in informal conversations
 - "Talkativeness" is not gender-specific